Statistics & Probability

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Measures of center**

Find the mean of $12, 15, 18, 9, 21$.

Sum $= 75$. Mean $= \frac{75}{5} = 15$.

2 **[Foundational]** • **Target: Measures of center**

Find the median of $4, 9, 2, 11, 7, 15$.

Sorted: $2, 4, 7, 9, 11, 15$. Median $= \frac{7+9}{2} = 8$.

3 **[Foundational]** • **Target: Measures of center**

Find the mode of $3, 7, 5, 7, 2, 7, 9$.

The value 7 appears most often, so the mode is 7.

4 **[Foundational]** • **Target: Measures of spread**

Find the range of the data set $22, 18, 35, 9, 27$.

Range $= 35 - 9 = 26$.

5 **[Foundational]** • **Target: Measures of spread**

Find the interquartile range (IQR) of $4, 7, 9, 12, 15, 18, 21$.

Lower half (excluding the median 12): $4, 7, 9$, so $Q_1 = 7$. Upper half: $15, 18, 21$, so $Q_3 = 18$. IQR $= 18 - 7 = 11$.

6 **[Foundational]** • **Target: Measures of spread**

A data set has a range of 40 and a minimum value of 15. Find the maximum value.

Maximum $= 15 + 40 = 55$.

7 **[Foundational]** • **Target: Basic probability**

A jar has 5 red, 3 blue, and 2 green marbles. Find $P(\text{green})$.

Total marbles $= 10$. $P(\text{green}) = \frac{2}{10} = \frac{1}{5}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Basic probability

A fair six-sided die is rolled once. Find $P(\text{even number})$.

Even outcomes: $\{2, 4, 6\}$. $P = \frac{3}{6} = \frac{1}{2}$.

9 [Intermediate] • Target: Basic probability

A spinner has 8 equal sections numbered 1 through 8. Find $P(\text{number greater than } 5)$.

Numbers greater than 5: $\{6, 7, 8\}$. $P = \frac{3}{8}$.

10 [Intermediate] • Target: Combined probability: AND / OR

A card is drawn from a standard 52-card deck. Find $P(\text{king or queen})$.

4 kings $+4$ queens $= 8$ favorable cards. $P = \frac{8}{52} = \frac{2}{13}$.

11 [Intermediate] • Target: Combined probability: AND / OR

Two independent events have $P(A) = 0.4$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.

$P(A \cap B) = 0.4 \times 0.5 = 0.20$.

12 [Intermediate] • Target: Combined probability: AND / OR

A bag has 6 yellow and 4 purple marbles. A marble is drawn, replaced, then a second is drawn. Find $P(\text{both yellow})$.

$P = \frac{6}{10} \times \frac{6}{10} = \frac{36}{100} = \frac{9}{25}$.

13 [Intermediate] • Target: Probability without replacement

A bag has 5 red and 7 blue marbles. Two marbles are drawn without replacement. Find $P(\text{both red})$.

$P = \frac{5}{12} \times \frac{4}{11} = \frac{20}{132} = \frac{5}{33}$.

14 [Intermediate] • Target: Probability without replacement

A committee of 2 is chosen at random, one at a time without replacement, from a group of 4 men and 6 women. Find $P(\text{both chosen are women})$.

$P = \frac{6}{10} \times \frac{5}{9} = \frac{30}{90} = \frac{1}{3}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Data displays • Cross-Lesson: 6.01 Geometry

The bar graph shows the results of rolling a fair die 40 times. Based on the experimental data, find the experimental probability of rolling a 4, and compare it to the theoretical probability.

Experimental $P(4) = \frac{9}{40} = 0.225$. Theoretical $P(4) = \frac{1}{6} \approx 0.167$. The experimental probability is somewhat higher than the theoretical probability, which is expected random variation in a finite sample.

16 [Challenging] • Target: Expected value • Cross-Lesson: 7.01 Trigonometry & Additional Topics

A game costs 2 dollars to play. A player wins 10 dollars with probability 0.1, and wins nothing otherwise. Find the expected net profit per play.

Expected winnings $= 0.1 \times 10 = 1$ dollar. Expected net profit $= 1 - 2 = -1$ dollar (an expected loss of 1 dollar per play).

17 [Challenging] • Target: Expected value • Cross-Lesson: 8.01 Exponentials & Logarithms

A spinner has 4 equal sections labeled 5, 10, 15, and 20 points. Find the expected value of one spin.

$E = \frac{5 + 10 + 15 + 20}{4} = \frac{50}{4} = 12.5$ points.

18 [Challenging] • Target: Effects of transformations on statistics • Cross-Lesson: 1.01 Heart of Algebra

A data set has mean 20. If every value in the set is increased by 5, find the new mean.

New mean $= 20 + 5 = 25$.

19 [Challenging] • Target: Effects of transformations on statistics • Cross-Lesson: 2.01 Quadratics & Polynomials

A data set has standard deviation 4. If every value is multiplied by 3, find the new standard deviation.

New standard deviation $= 3 \times 4 = 12$.

20 [Challenging] • Target: A well-designed word problem: a survey of students • Cross-Lesson: 3.01 Functions & Graphs

This question has three parts. A researcher surveys 200 students about whether they play a sport and whether they play a musical instrument. Of the 200 students, 120 play a sport, 80 play an instrument, and 30 do both. (a) Find how many students do neither activity. (b) Find the probability that a randomly chosen student plays a sport or an instrument (or both). (c) Given that a randomly selected student plays a sport, find the probability that they also play an instrument.

(a) By inclusion-exclusion, the number doing at least one activity is $120 + 80 - 30 = 170$. Neither $= 200 - 170 = 30$. (b) $P(\text{sport or instrument}) = \frac{170}{200} = 0.85$. (c) $P(\text{instrument} \mid \text{sport}) = \frac{P(\text{both})}{P(\text{sport})} = \frac{30}{120} = \frac{1}{4}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Measures of center

Find the mean of $6, 50, 46, 36, 7$.

a. 29 ✓

b. 31

c. 145

d. 27

Sum $= 145$. Mean $= \frac{145}{5} = 29$.

22 [Foundational] • Target: Measures of spread

Find the mean of $21, 18, 14, 35, 2$.

a. 90

b. 16

c. 18 ✓

d. 20

Sum $= 90$. Mean $= \frac{90}{5} = 18$.

23 [Foundational] • Target: Basic probability

Find the mean of $23, 12, 44, 5, 26$.

a. 24

b. 20

c. 110

d. 22 ✓

Sum $= 110$. Mean $= \frac{110}{5} = 22$.

24 [Foundational] • Target: Combined probability: AND / OR

Find the mean of $4, 31, 3, 26, 1$.

a. 13 ✓

b. 11

c. 65

d. 15

Sum $= 65$. Mean $= \frac{65}{5} = 13$.

25 [Foundational] • Target: Probability without replacement

Find the mean of $9, 21, 19, 19, 22$.

a. 18 ✓

b. 16

c. 20

d. 90

Sum $= 90$. Mean $= \frac{90}{5} = 18$.

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26 [Foundational] • Target: Data displays

Find the mean of $50, 23, 27, 33, 12$.

a. 145

b. 31

c. 29 ✓

d. 27

Sum $= 145$. Mean $= \frac{145}{5} = 29$.

27 [Foundational] • Target: Expected value

Find the mean of $14, 29, 33, 44, 30$.

a. 28

b. 32

c. 30 ✓

d. 150

Sum $= 150$. Mean $= \frac{150}{5} = 30$.

28 [Foundational] • Target: Effects of transformations on statistics

A jar has 11 yellow, 6 red, and 6 green marbles. Find $P(\text{red})$.

a. $\frac{6}{23}$ ✓

b. $\frac{8}{23}$

c. $\frac{11}{23}$

d. $\frac{9}{23}$

Total marbles $= 23$. $P(\text{red}) = \frac{6}{23}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: a survey of students

A jar has 4 red, 8 purple, and 4 green marbles. Find $P(\text{red})$.

a. $\frac{7}{16}$

b. $\frac{3}{8}$

c. $\frac{1}{4}$ ✓

d. $\frac{1}{2}$

Total marbles $= 16$. $P(\text{red}) = \frac{4}{16} = \frac{1}{4}$.

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30 [Intermediate] • Target: Summary Statistics & Outliers

A jar has 9 red, 2 green, and 10 blue marbles. Find $P(\text{red})$.

a. $\frac{2}{21}$

b. $\frac{10}{21}$

c. $\frac{11}{21}$

d. $\frac{3}{7}$ ✓

Total marbles $= 21$. $P(\text{red}) = \frac{9}{21} = \frac{3}{7}$.

31 [Intermediate] • Target: Normal Distributions & Standard Deviation

A jar has 2 red, 5 purple, and 1 orange marbles. Find $P(\text{red})$.

a. $\frac{1}{8}$

b. $\frac{1}{2}$

c. $\frac{5}{8}$

d. $\frac{1}{4}$ ✓

Total marbles $= 8$. $P(\text{red}) = \frac{2}{8} = \frac{1}{4}$.

32 [Intermediate] • Target: Compound & Conditional Probability

A jar has 10 red, 8 purple, and 9 blue marbles. Find $P(\text{red})$.

a. $\frac{1}{3}$

b. $\frac{8}{27}$

c. $\frac{10}{27}$ ✓

d. $\frac{4}{9}$

Total marbles $= 27$. $P(\text{red}) = \frac{10}{27}$.

33 [Intermediate] • Target: Permutations & Combinations

A jar has 12 green, 12 orange, and 2 blue marbles. Find $P(\text{blue})$.

a. $\frac{5}{26}$

b. $\frac{1}{13}$ ✓

c. $\frac{2}{13}$

d. $\frac{6}{13}$

Total marbles $= 26$. $P(\text{blue}) = \frac{2}{26} = \frac{1}{13}$.

34 [Intermediate] • Target: Measures of center

A jar has 9 blue, 10 orange, and 3 red marbles. Find $P(\text{blue})$.

a. $\frac{5}{11}$

b. $\frac{9}{22}$ ✓

c. $\frac{3}{22}$

d. $\frac{1}{2}$

Total marbles $= 22$. $P(\text{blue}) = \frac{9}{22}$.

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35 [Intermediate] • Target: Measures of spread

Two independent events have $P(A) = 0.1$ and $P(B) = 0.7$. Find $P(A \text{ and } B)$.

a. 0.27

b. 0.17

c. 0.12

d. 0.07 ✓

$P(A \cap B) = 0.1 \times 0.7 = 0.07$.

36 [Intermediate] • Target: Basic probability

Two independent events have $P(A) = 0.8$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.

a. 0.40 ✓

b. 0.30

c. 0.50

d. 0.60

$P(A \cap B) = 0.8 \times 0.5 = 0.40$.

37 [Intermediate] • Target: Combined probability: AND / OR

A bag has 7 red and 9 blue marbles. Two marbles are drawn without replacement. Find $P(\text{both red})$.

a. $\frac{7}{16}$

b. $\frac{21}{40}$

c. $\frac{3}{10}$

d. $\frac{7}{40}$ ✓

$P = \frac{7}{16} \times \frac{6}{15} = \frac{42}{240} = \frac{7}{40}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Probability without replacement • Cross-Lesson: 1.01 Heart of Algebra

A bag has 7 red and 4 blue marbles. Two marbles are drawn without replacement. Find $P(\text{both red})$.

a. $\frac{21}{55}$ ✓

b. $\frac{28}{55}$

c. $\frac{7}{11}$

d. $\frac{6}{55}$

$P = \frac{7}{11} \times \frac{6}{10} = \frac{42}{110} = \frac{21}{55}$.

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39 [Challenging] • Target: Data displays • Cross-Lesson: 2.01 Quadratics & Polynomials

A spinner has 4 equal sections labeled 10, 20, 25, and 40 points. Find the expected value of one spin.

a. 26.75

b. 24.75

c. 23.75 ✓

d. 22.75

$E = \frac{10 + 20 + 25 + 40}{4} = \frac{95}{4} = 23.75$ points.

40 [Challenging] • Target: Expected value • Cross-Lesson: 3.01 Functions & Graphs

A spinner has 4 equal sections labeled 15, 20, 30, and 40 points. Find the expected value of one spin.

a. 27.25

b. 26.25 ✓

c. 29.25

d. 25.25

$E = \frac{15 + 20 + 30 + 40}{4} = \frac{105}{4} = 26.25$ points.

41 [Challenging] • Target: Summary Statistics & Outliers • Cross-Lesson: 4.01 Ratios, Rates & Data

The mean of five numbers is 24. When a sixth number is added, the mean becomes 26. What is the sixth number?

a. 32

b. 34

c. 36 ✓

d. 38

Sum of 5 numbers $= 5 \times 24 = 120$. Sum of 6 numbers $= 6 \times 26 = 156$. Sixth number $= 156 - 120 = 36$.

42 [Challenging] • Target: Summary Statistics & Outliers • Cross-Lesson: 6.01 Geometry

An outlier is defined as any data point greater than $Q_3 + 1.5 \times \text{IQR}$. If $Q_1 = 20$ and $Q_3 = 35$, which value is considered an outlier?

a. 50

b. 55

c. 58

d. 60 ✓

$\text{IQR} = 35 - 20 = 15$. Upper threshold $= 35 + 1.5(15) = 35 + 22.5 = 57.5$. Since $60 > 57.5$, 60 is an outlier.

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43 [Challenging] • Target: Normal Distributions & Standard Deviation • Cross-Lesson: 7.01 Trigonometry & Additional Topics

Which change will increase the standard deviation of a dataset?

a. Adding 10 to all data points

b. **Multiplying all data points by 2** ✓

c. Subtracting the mean from each data point

d. Dividing all data points by 3

Multiplying by 2 doubles the dispersion and doubles the standard deviation. Adding/subtracting a constant shifts the data without affecting standard deviation.

44 [Challenging] • Target: Compound & Conditional Probability • Cross-Lesson: 8.01 Exponentials & Logarithms

If $P(A) = 0.5$, $P(B) = 0.6$, and $P(A \cup B) = 0.8$, what is $P(A \cap B)$?

a. 0.1

b. 0.2

c. 0.3 ✓

d. 0.4

$P(A \cup B) = P(A) + P(B) - P(A \cap B) \Rightarrow 0.8 = 0.5 + 0.6 - P(A \cap B) \Rightarrow P(A \cap B) = 1.1 - 0.8 = 0.3$.

45 [Challenging] • Target: Permutations & Combinations • Cross-Lesson: 1.01 Heart of Algebra

How many distinct 3-person executive committees (President, Vice-President, Treasurer) can be elected from a club of 10 members?

a. 120

b. 240

c. 720 ✓

d. 1000

Since roles are distinct, order matters (permutation): $P(10, 3) = 10 \times 9 \times 8 = 720$.

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